

The Ephemeris



June 2024

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June 2024 - September 2024 Events

Schedule subject to change
Please check our website for up
to date information.

Board & General Meetings

Saturday

6/22, 7/13, 8/17, 9/14

Board Meetings: 6 - 7:30pm

General Meetings: 8:00 -
9:30pm

Solar Sunday (2:00 - 4:00pm)

Sunday 6/2, 7/7, 8/4, 9/8

In-Town Star Party (7:00- 9:00pm)

Friday 6/7, 7/5, 8/2, 9/27

RCDO Starry Nights Star Party (7:00 - 9:00pm)

Saturday 6/1, 6/29, 7/27, 8/31

Imaging SIG (online)

Tuesday 6/18, 7/16, 8/20, 9/17

Binocular Star Gazing (at Coyote Valley OSP)

Saturday 6/1, 6/29

*Please refer to the SJAA
Website home page
(www.sjaa.net) and click
on any "Upcoming Event"
highlighted in Blue.*

INSIDE THIS ISSUE

Astrophoto	2
Introducing Director 2	3
Introducing Program Leaders	4
Observation Report	5
Solar Eclipse Trip	6
Astrophotos	7
Marianne's Meanderings	11
Membership Form	12
Board Meeting Minutes	13
Contact Us	14

Swaroop Shere Photography

Total Solar Eclipse 2024 Prominences
Photo by SJAA Member: Swaroop Shere
April 8, 2024
Location: Mazatlán Mexico



M 16 and NGC 6604

Photo by: Rob Pfile

May 9, 2024

Introducing Director 2

Credit: Radhika Gupta



Since my earliest memories, I've been captivated by the wonders of Astronomy. Armed with a Bachelor's degree in Physics and Mathematics, I've navigated a career as a software engineer and technical program manager. However, my heart truly belongs to the cosmos. Whether it's gazing at the night sky, reading Astronomy literature, tuning into celestial podcasts, or engaging in conversations with fellow enthusiasts, my passion for the stars burns brighter than ever. Going forward, I'm committed to exploring the universe with even greater rigor and enthusiasm!

Introducing Astronomy 101 and Library Program Leaders

Introducing Carl-Erik Svensson— Astro 101 Program Leader



I have been a member of SJAA since 2017, when I got my first telescope at the tender age of 33, after a lifetime of casual interest in astronomy, and all things space. I continue to use my Celestron NexStar 8SE at various public outreach events, including school star parties and Starry Nights at RCDO. I've since acquired two additional refractors, both Stellarvue brand: one 80mm, and another 130mm. Each of these are part of separate imaging rigs, with which I've had great fun (and frustration) exploring the wonderful world of astrophotography. I recently viewed my first total solar eclipse just this year, for which I built a pair of binoculars with solar filters. If that weren't enough, I'm planning a roll-off-roof observatory build under a dark (ish) site in the Sierras. I'm looking forward to continuing the tradition of giving introductory astronomy talks to members and the public alike, as well as expanding the Astro 101 role to online forums and beyond. I also recently started helping with The Ephemeris newsletter (like this one!). My images are at <https://www.astrobin.com/users/csvensson/>.

Introducing David Kershaw — Librarian



David grew up in San Diego where he also went to school and studied Physics. (As a kid there were fond memories of family trips and school field trips to Palomar Observatory.) After that one of his first jobs was as a Metrology Engineer. This was followed by Process Engineer, then a Network Engineer with the factory automation department and most recently as a Wireless Network Engineer. At night David likes to look up at the sky and play with radios. An early student job was to work in a library, so librarian seemed like a good volunteer position.

Observation Report Total Solar Eclipse

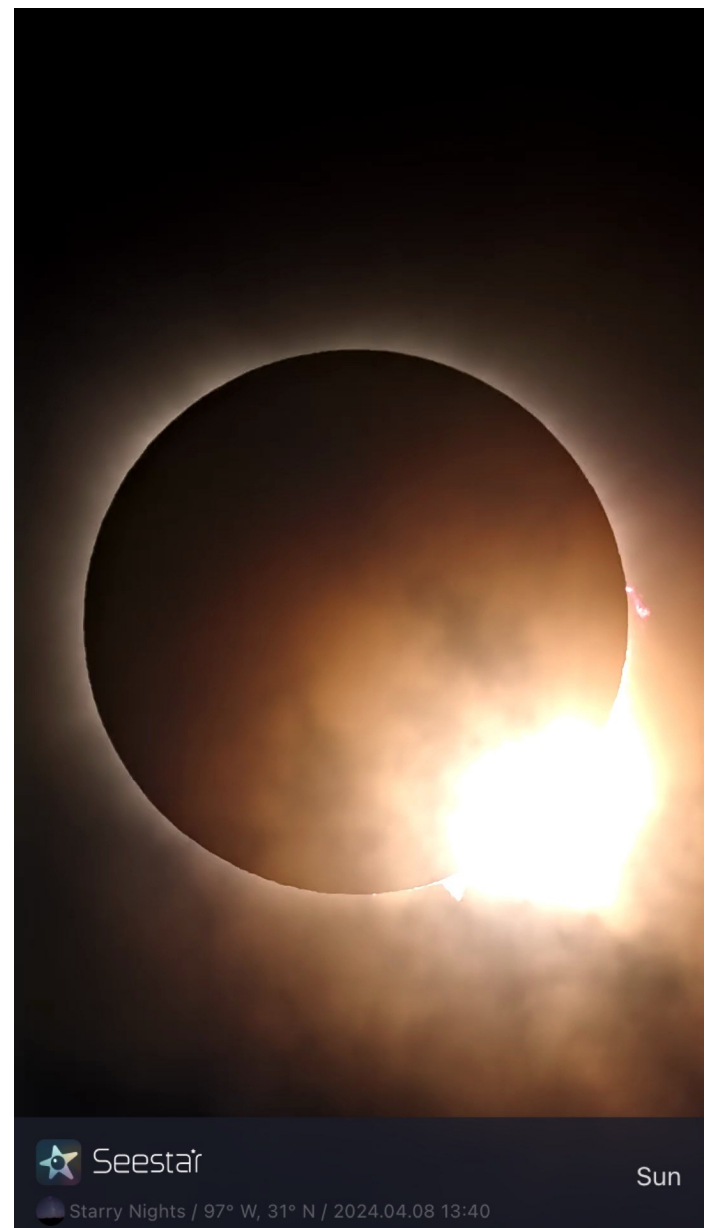
Credit: Joseph Fragola

The weather forecast for viewing the total solar eclipse did not look promising one week out from the big day. After driving our RV 1,749 miles from San Jose, my wife and I arrived at Copperas Cove, Texas around 1 PM on Sunday, April 7. On Eclipse Day we woke up to mostly cloudy skies. But as the morning progressed there were significant breaks in the clouds with large patches of blue sky. First contact from our location was at 12:18 PM local time. Sky conditions continued to be favorable as the eclipse started. But the weather forecast for our location indicated incoming clouds with thunderstorms developing in the afternoon. So we knew it was only a matter of time before clouds would arrive.

Sky conditions remained favorable for viewing up until about 5 - 10 minutes before totality. At that point, our eclipse viewing experience was reduced to watching our surroundings get darker as we got closer to totality.

Fortunately, just after totality the clouds began to dissipate enough that we were able to view the eclipsed Sun, although the full extent of the corona was muted due to the clouds. As totality progressed, the sky conditions continued to improve. We were able to continue watching the partial phases as the Moon exited the face of the Sun up until 5 minutes before the conclusion of the eclipse when it became too cloudy.

I used my Seestar S50 to view/image the Sun during the eclipse.



Marianne's Meanderings

Credit: Marianne Damon

After unpacking from our 2017 Madras, Oregon trip I looked up the path for the next “big one”. “Hmmm, my daughter’s in-laws live smack under the center line for the 2024 TSE! I have seven years to cajole them into a family gathering...” My scheming worked, plans were made, and the day finally came when we set off on a road trip to China Spring, just northwest of Waco. We stopped at the Grand Canyon, Monument Valley, and Taos. Each day I watched the waning crescent Moon grow thinner. By Saturday night my daughter and family had arrived from Boston and on Sunday we set up our gear for a “dry run.” I had my BT-70 with filters, 10x50 binoculars with filters, the Celestron EclipseSmart 10x42’s, and framed eclipse glasses for everyone. My son-in-law had his spotting scope, binoculars, and camera all with solar filters purchased well beforehand after consulting with me. Clouds came and went, but we had decent overhead views. My daughter, who had been rather ho-hum about seeing a total eclipse, became more interested. My 9-year-old grandson helped me set up and stuck by my side the entire time. He walked me through what would be happening the next day, thanks to a NASA for kids’ video he had watched, from his Planetary Society quarterly kits he receives, and from a Curious George cartoon! My five-year-old granddaughter was busy with all her little ponies. Our pre-viewing session was a success. However, the big topic, amongst us and on my online and text groups, was CLOUD ANXIETY. Stay? Go? Go how far? I was content to stay put and experience with family whatever happened, although my hosts were worried that Texas would “let us down.” Monday morning was cloudy. We set up chairs, brought out the ice chest filled with “free drinks” and a wagon with “free snacks.” We spread out a white sheet in case we could “catch” the shadow bands. Just before noon the clouds started thinning and moving, and we had clear high views with just a few episodes of obscuration. First Contact was right on schedule and everyone had fun comparing views through the different equipment as the eclipse progressed. We watched the sunspots get “gobbled up.” I made sure my grandson was at the BT for Bailey’s Beads and the Diamond Ring. And Totality? Everyone felt that sense of awe. We marveled at the prominences. We spotted Venus and Jupiter. We felt that deep “hush.” I was so thankful for that four minutes and 23 seconds! We witnessed how quickly that final arc merged into Totality and then grew back again. And in reverse, we watched...and the Moon “spit out” the sunspot. We didn’t see shadow bands, but spent time playing with the colander and found that even a Cheez-It hole produced an eclipse view if held at the right distance! We kept watching until the final, final moment when the Moon left the Sun. After stowing away our gear, I got out my Planet Walk Activity. We placed the Sun where we had been observing. Every three big steps equal an Astronomical Unit, and off we went until finally, up the road by the barn, we placed Pluto and the Kuiper Belt, and speculated how much farther out the Oort Cloud would be. By then we were hot and the clouds were rolling in and we made it back inside to await the storm. For my first TSE in 1979 in Wolf Point, Montana I was all by myself...in 2017 Dave and I were with SJAA friends and lots of campers...this time we were sharing the event with family, and it felt so special. I feel so fortunate!





Image: Abell 2151 The Hercules Cluster
By: Hy Murveit
Dates: April 20, 2024
Location: Home, Silicon Valley, California, United States
Telescope: GSO 10" f/8 Ritchey-Chretien Truss Tube
Camera: ZWO ASI1600MM Pro
Mount: Astro-Physics 1100GTO
Software: Adobe Lightroom Classic, Adobe Photoshop, KDEdu KStars, Pleiades Astrophoto PixInsight

Astrobin Link: <https://www.astrobin.com/4084h4/>

SJAA Member Astrophoto Gallery



Image: NGC 5005 and NGC 5033
By: Francesco Meschia
Dates: May 5-12, 2024
Location: Home, Mountain View, CA
Telescope: Astro-Tech AT130EDT
Camera ZWO ASI294MM Pro
Mount iOptron iEQ45 Pro
Software Adobe Photoshop, Cranfield / Payne Generalized Hyperbolic Stretching (GHS) Module/Script, John Murphy NormalizeScaleGradient Script, Pleiades Astrophoto PixInsight, ·Russell Croman Astrophotography BlurXTerminator, Russell Croman Astrophotography NoiseXTerminator, ·Russell Croman Astrophotography StarXTerminator, ·Stefan Berg Nighttime Imaging 'N' Astronomy (N.I.N.A. / NINA)

Astrobin Link <https://www.astrobin.com/zn5vzf/>

SJAA Member Astrophoto Gallery



Image: NGC 4565 The Needle Galaxy
By: John Sims
Dates: April 30—May 1, 2024
Location: Mt. Umunhum Trailhead, San Jose, CA, United States
Telescope: Stellarvue SVX102T
Camera: ZWO ASI2600MC Pro
Mount: ZWO AM5
Software Phase One Capture One Pro, ·Pleiades Astrophoto PixInsight, ·Russell Croman Astrophotography BlurXTerminator, ·Russell Croman Astrophotography NoiseXTerminator, ·Russell Croman Astrophotography StarXTerminator

Astrobin Link: <https://www.astrobin.com/wnnjfa/D/>



Image: The Great Orion Nebula
By: Nikola Nikolov
Dates: April 2, 2024
Location: Adelaida Vineyards, Paso Robles, CA
Telescope: Askar FMA180
Camera: Canon EOS 700D / Rebel T5i / Kiss X7i
Mount: iOptron SkyGuider Pro
Software: Aries Productions Astro Pixel Processor (APP), O'Telescope BackyardEOS, Pleiades Astrophoto Pix-Insight , · Russell Croman Astrophotography NoiseXTerminator

Astrobin link: <https://www.astrobin.com/3gqa7e/>

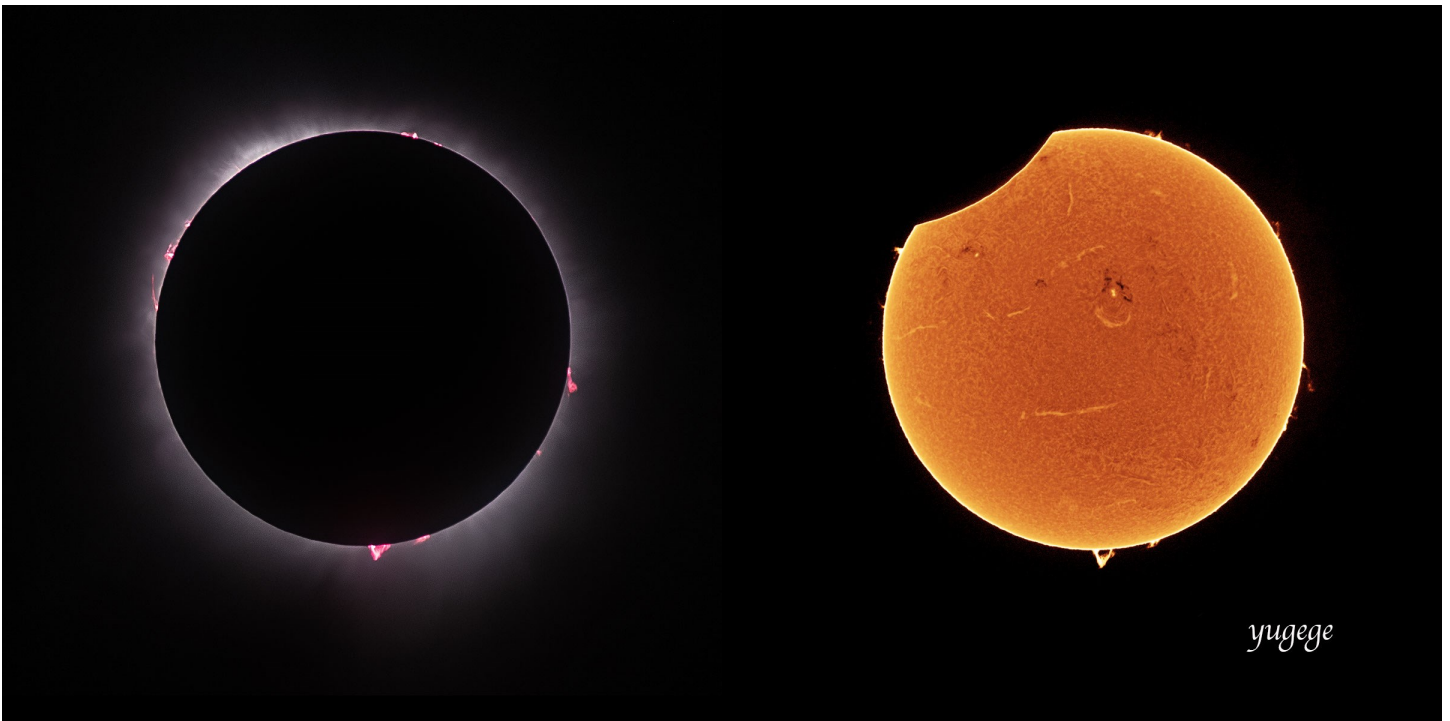
Solar Eclipse Trip

Credit: Yu Li

I drove 1,700 miles with my equipment to Killeen, Texas, and met friends and family who flew there too. Because of the bad weather forecasts, we planned to relocate to various locations but ultimately drove to Grannis, Arkansas on Apr 7th, for clear sky forecast. However, as the eclipse approached in the morning of Apr 8th, thick clouds gathered, the sun was nowhere to be seen. The situation looked hopeless at that time. 30 minutes before totality, we finally decided to move south, leaving all the heavy telescopes and sophisticated photograph plan behind. Like a miracle, we found a patch of blue sky 8 minutes before the totality, allowing us to witness the total eclipse on the roadside. Despite I had to manually press the shutters to take the pictures, I managed to capture memorable photos with a single tripod and camera. Upon returning to the campsite, we learned the clouds had cleared just 5 minutes before totality, and people also watched the eclipse there.

Attached are two pictures. The left sun in the first picture is a composition of C2 and C3 to show the prominence. the right sun was taken by a Lunt 60mm H-alpha solar telescope after I returned to the campsite during the partial eclipse. It is interesting to compare the two.

The 2nd picture was a stack of 9 pictures to show the corona of the sun.



San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with online Ephemeris

☐ **\$30** Regular Membership with hardcopy Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____

From the Board of Directors

Announcements

- ◇ Aroon Nair, Swami Nigam, and Rob Jaworski were reelected as President, Secretary and Treasurer. The VP position is open.
- ◇ Current SJAA Members can benefit from signing up for the “SJ Astronomy” group on Meetup (<https://www.meetup.com/sj-astronomy/>) as well as the “SJAA Observers” Google group (<https://groups.google.com/g/sjaa-observers>).
- ◇ New Librarian is David Kershaw
- ◇ New Astronomy 101 leader is Carl-Eric Svensson
- ◇ Open positions: Ephemeris Production Editor, Ephemeris Content Editor, and Mendoza Ranch

Board Meeting Excerpts

March 23, 2024

In attendance

Aroon, Swami, Rob, Vini, Ken, Glenn, Peter, Wolf, Marianne

Absent: Chris

Guests: Leo L, Larry Z, Walter T, Frances. Nancy L, Emi N, Nikola, David Kershaw.

Officer Elections

SJAA Officer elections were to be held March 16-23. Aroon Nair, Swami Nigam, and Rob Jaworski were reelected as President, Secretary and Treasurer. The Vice President position is vacant.

Swap Meet

Ken will coordinate the planning and execution of the next SJAA swap meet.

April 20, 2024

In attendance

Aroon, Ken, Wolf, Swami, Rob J, Peter, Marianne, Vini

Absent: None

Guests: Emi, Carl S, Larry, David K

Money Request for Solar: Power Stations and Drawing Prizes

Wolf asked for \$1600 for the purchase of power stations and drawing prizes. Rob made the motion and it was seconded by Marianne all approved. \$3000 (including taxes) to purchase the equipment requested above by the working group. Glenn seconded.

All approved, no abstentions or objections. Motion carried.

May 18, 2024

In attendance

Aroon, Ken, Wolf, Swami, Rob J, Glenn, Marianne, Vini

Absent: Peter, Radhika

Guests: David K, Michael M, Joe F, Frances, Walter, Emi N, Leo L

Resignations and Open Positions

Emi communicated her decision to step down from the Ephemeris Production Editor post, after the June edition. Since Abishek has also resigned, we will have no Ephemeris Editors, so need to quickly find replacements. Bob and Natti Thompson have indicated they will be stepping down from the Mendoza Ranch program leaders.

Lunt 80 Blocking Filter Repair

Motion to approve Wolf's request for funds to cover the costs of the Lunt 80 Blocking Filter repair; Marianne moved that no more than \$400 be appropriated, seconded by Glenn, all approved.

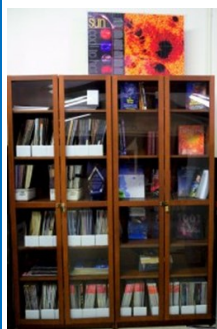
Swap Meet

Agreement to hold a Swap Meet with the tentative date of September 15, 2024.

Editor's Note 1: On March 2024 issue page 2 Richie's last name is spelled wrong; it should be Senegal instead of Senegal.

Editor's Note 2: I have decided to step down from the Ephemeris Editor position. This is my final issue as an editor. Good luck to the new team! Thanks to everyone who has been working with me and helping improve the Ephemeris!

SJAA Library



SJAA offers another wonderful resource: a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program, is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Larry Zurbick, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope. Please note that this program is for members only!

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Hogue Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Jonathan Lawton, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions. Please note that this program is for members only!

The Quick Start sessions are generally held monthly. The next Quick Start session during this time period is end of June.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Hogue Park. This is a regular, monthly session, run by Carl-Erik Svensson, each

with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Hy Murveit, the Imaging SIG meets roughly every month at Hogue Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group, this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Hogue Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

SJAA Contacts

President:	Aroon Nair
Vice President:	Open
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Swami Nigam
Director:	Vini Carter
Director:	Peter Melhus
Director:	Radhika Gupta
Director:	Wolf Witt
Director:	Glenn Newell
Director:	Marianne Damon
Director:	Ken Miura
Ephemeris	
Content Editor:	Abishek Ramasubramanian
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Larry Zurbick
Hands on Imaging:	Glenn Newell
Imaging SIG:	Hy Murveit
Intro to the Night Sky:	Carl-Erik Svensson
In-Town Star Party:	Muditha Kanchana
Library:	David Kershaw
Loaner Program:	Muditha Kanchana
Memberships:	Kaushal Gangakhedkar
Publicity:	Rob Jaworski
Donations:	Rich Gregor
Quick START:	Jonathan Lawton
Solar:	Wolf Witt
Starry Nights:	Joseph Fragola
School Events:	Jon Anderson
Refreshments:	Marianne Damon
Social Media Manager:	Beth Johnson
Speakers:	Leo Lipkin
Webmaster:	Satish Vellanki
Volunteer:	Nancy Lieu
E-mails:	http://www.sjaa.net/contact

<http://www.sjaa.net/programs/school-star>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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